

# Local Nonlinear Projection-Based Reduced-Order Models

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**Problem:** 2D Burgers, mesh  $250 \times 250$ .

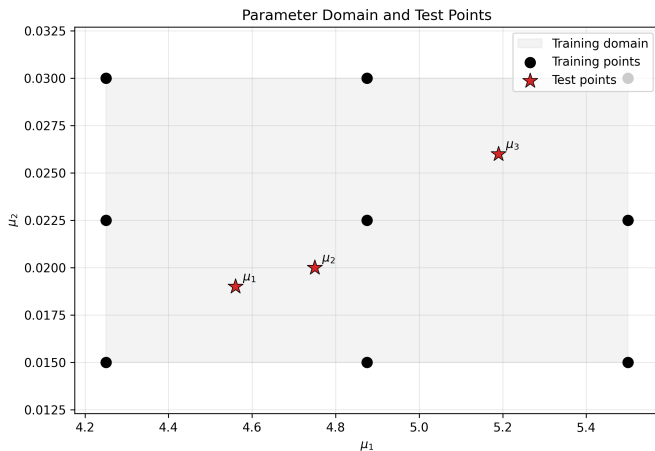
**Test points:**

$$\mu_1 = (4.56, 0.019), \quad \mu_2 = (4.75, 0.020), \quad \mu_3 = (5.19, 0.026).$$

**Methods included:**

- ▶ Global HPROM (linear POD)
- ▶ Global HQPROM (quadratic manifold)
- ▶ Global HPROM-GPR
- ▶ Local HPROM
- ▶ Local HQPROM
- ▶ Local HPROM-GPR

## Parameter Domain and Test Points



## Global ROM Storyline: LSPG + Gauss-Newton

Given the implicit HDM residual  $\mathbf{r}(\mathbf{u}) = \mathbf{0}$ , define a manifold

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}).$$

LSPG solves at each time step:

$$\mathbf{q}^{\star} = \arg \min_{\mathbf{q}} \frac{1}{2} \|\mathbf{r}(\mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}))\|_2^2.$$

With Gauss-Newton (current iterate  $\mathbf{q}^{(k)}$ ), the increment  $\Delta \mathbf{q}$  is obtained from

$$(\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{J} \boldsymbol{\Psi}) \Delta \mathbf{q} = -\boldsymbol{\Psi}^T \mathbf{J}^T \mathbf{r},$$

where

$$\boldsymbol{\Psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}}, \quad \mathbf{J} = \frac{\partial \mathbf{r}}{\partial \mathbf{u}}, \quad \mathbf{q}^{(k+1)} = \mathbf{q}^{(k)} + \Delta \mathbf{q}.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q}, \quad \mathbf{V} \in \mathbb{R}^{N \times n}, \quad n = 96.$$

Hence

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q}, \quad \boldsymbol{\psi} = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V}.$$

Gauss-Newton system:

$$(\mathbf{V}^T \mathbf{J}^T \mathbf{J} \mathbf{V}) \Delta \mathbf{q} = -\mathbf{V}^T \mathbf{J}^T \mathbf{r}.$$

Quadratic-manifold ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}), \quad n = 39.$$

Thus

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}\mathbf{q} + \mathbf{H}(\mathbf{q} \otimes \mathbf{q}),$$

and the tangent depends on the current reduced state:

$$\boldsymbol{\Psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \frac{\partial}{\partial \mathbf{q}}[\mathbf{H}(\mathbf{q} \otimes \mathbf{q})].$$

Split basis and reduced coordinates:

$$\mathbf{V}_{\text{tot}} = [\mathbf{V}, \bar{\mathbf{V}}], \quad \mathbf{q}_{\text{tot}} = \begin{bmatrix} \mathbf{q} \\ \bar{\mathbf{q}} \end{bmatrix}.$$

$$\bar{\mathbf{q}} = \mathcal{N}_{\text{GPR}}(\mathbf{q}), \quad n = 20, \bar{n} = 131.$$

State ansatz:

$$\mathbf{u} \approx \mathbf{u}_{\text{ref}} + \tilde{\mathbf{u}}(\mathbf{q}),$$

$$\tilde{\mathbf{u}}(\mathbf{q}) = \mathbf{V}_{\text{tot}} \mathbf{q}_{\text{tot}} = \mathbf{V} \mathbf{q} + \bar{\mathbf{V}} \mathcal{N}_{\text{GPR}}(\mathbf{q}).$$

Effective tangent for Gauss-Newton:

$$\boldsymbol{\psi}(\mathbf{q}) = \frac{\partial \tilde{\mathbf{u}}}{\partial \mathbf{q}} = \mathbf{V} + \bar{\mathbf{V}} \frac{\partial \mathcal{N}_{\text{GPR}}}{\partial \mathbf{q}}.$$

## Local ROMs: Piecewise Manifolds

Partition the training snapshots into clusters  $k = 1, \dots, N_c$  and build one local trial manifold per cluster:

$$\mathbf{u} \approx \Phi_k(\mathbf{q}_k).$$

Examples used in this work:

$$\Phi_k^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q},$$

$$\Phi_k^{\text{quad}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \mathbf{H}_k(\mathbf{q} \otimes \mathbf{q}),$$

$$\Phi_k^{\text{gpr}}(\mathbf{q}) = \mathbf{u}_{0,k} + \mathbf{V}_k \mathbf{q} + \bar{\mathbf{V}}_k \mathcal{N}_{\text{GPR},k}(\mathbf{q}).$$

### Online workflow

1. Select the active cluster.
2. If the cluster changes, transfer the current state to the new local manifold.
3. Solve the local LSPG problem in the active manifold.

## Cluster Selection: Final Form (Linear)

Shared nearest-centroid selector (used in all local models):

$$k^+ = \arg \min_{\ell=1,\dots,N_c} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2, \quad \mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

For the linear local manifold  $\Phi_{k^-}^{\text{lin}}(\mathbf{q}) = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-}\mathbf{q}$ :

$$k^+ = \arg \min_{\ell=1,\dots,N_c} S_{k^-, \ell}^{\text{lin}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{lin}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + d_{k^-, \ell},$$

with offline-precomputed terms

$$\mathbf{g}_{k^-, \ell} := \mathbf{V}_{k^-}^T (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \quad d_{k^-, \ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2.$$

## Cluster Selection: Final Form (HQPROM and HPROM-GPR)

### HQPROM selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{quad}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{quad}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) + d_{k^-, \ell},$$

$$\mathbf{m}_{k^-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell})^T \mathbf{H}_{k^-} (\mathbf{q} \otimes \mathbf{q}).$$

### HPROM-GPR selector

$$k^+ = \arg \min_{\ell} S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}_{k^-}),$$

$$S_{k^-, \ell}^{\text{gpr}}(\mathbf{q}) = 2 \mathbf{g}_{k^-, \ell}^T \mathbf{q} + 2 \bar{\mathbf{g}}_{k^-, \ell}^T \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}) + d_{k^-, \ell}.$$

$$\bar{\mathbf{q}}_{k^-} := \mathcal{N}_{\text{GPR}, k^-}(\mathbf{q}_{k^-}), \quad \bar{\mathbf{g}}_{k^-, \ell} := \bar{\mathbf{V}}_{k^-}^T (\mathbf{u}_{0, k^-} - \mathbf{u}_{c, \ell}).$$

## Cluster Change: Linear and Nonlinear Transfers

$k^+ = k^- \Rightarrow$  no transfer,  $k^+ \neq k^- \Rightarrow$  solve transfer problem.

Transferred state:

$$\mathbf{u}^s = \Phi_{k^-}(\mathbf{q}_{k^-}).$$

**Linear local target**

$$\mathbf{q}_{k^+}^{\text{lin}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - (\mathbf{u}_{0,k^+} + \mathbf{V}_{k^+} \mathbf{q}) \right\|_2^2,$$

$$\text{if } \mathbf{V}_{k^+}^T \mathbf{V}_{k^+} = \mathbf{I}, \quad \mathbf{q}_{k^+}^{\text{lin}} = \mathbf{V}_{k^+}^T (\mathbf{u}^s - \mathbf{u}_{0,k^+}).$$

**Nonlinear local targets (HQPROM / HPROM-GPR)**

$$\mathbf{q}_{k^+}^{\text{nl}} = \arg \min_{\mathbf{q}} \left\| \mathbf{u}^s - \Phi_{k^+}^{\text{nl}}(\mathbf{q}) \right\|_2^2.$$

## Cluster Change: Nonlinear Least Squares

For nonlinear local manifolds, define

$$\mathbf{r}_{\text{tr}}(\mathbf{q}) := \Phi_{k^+}(\mathbf{q}) - \mathbf{u}^s, \quad \mathbf{J}_{\Phi}(\mathbf{q}) := \frac{\partial \Phi_{k^+}}{\partial \mathbf{q}}.$$

Gauss–Newton at iterate  $\mathbf{q}^{(j)}$ :

$$(\mathbf{J}_{\Phi}^T \mathbf{J}_{\Phi}) \Delta \mathbf{q} = -\mathbf{J}_{\Phi}^T \mathbf{r}_{\text{tr}}(\mathbf{q}^{(j)}), \quad \mathbf{q}^{(j+1)} = \mathbf{q}^{(j)} + \Delta \mathbf{q}.$$

## Online Performance ( $250 \times 250$ ): Avg/Max Error over 3 Test Points

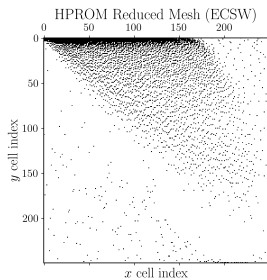
| Computational model           | $n$ ( $N$ for HDM) | $\bar{n}$ | $N_e$  | avg $\mathbb{RE}_{2,u}$ (%) | max $\mathbb{RE}_{2,u}$ (%) | Wall-clock time (min) | Speedup |
|-------------------------------|--------------------|-----------|--------|-----------------------------|-----------------------------|-----------------------|---------|
| HDM                           | 125 000            | –         | 62 500 | –                           | –                           | 12.517                | –       |
| HPROM                         | 96                 | –         | 4 241  | 1.030                       | 1.096                       | 0.366                 | 34.2    |
| HQPPROM                       | 39                 | –         | 3 460  | 0.818                       | 0.861                       | 0.641                 | 19.5    |
| HPROM-GPR                     | 20                 | 131       | 1 799  | 0.666                       | 0.785                       | 0.236                 | 53.1    |
| Local HPROM ( $N_c = 3$ )     | 35–48              | –         | 3 405  | 0.897                       | 0.933                       | 0.154                 | 81.4    |
| Local HQPPROM ( $N_c = 3$ )   | 21–27              | –         | 2 152  | 0.888                       | 1.025                       | 0.215                 | 58.3    |
| Local HPROM-GPR ( $N_c = 3$ ) | 9                  | 60–97     | 811    | 0.714                       | 0.786                       | 0.113                 | 110.5   |
| Local HPROM ( $N_c = 5$ )     | 23–31              | –         | 2 384  | 0.915                       | 0.943                       | 0.095                 | 132.0   |
| Local HQPPROM ( $N_c = 5$ )   | 15–20              | –         | 1 593  | 0.760                       | 0.878                       | 0.128                 | 97.9    |
| Local HPROM-GPR ( $N_c = 5$ ) | 7                  | 41–68     | 631    | 0.673                       | 0.880                       | 0.094                 | 133.2   |
| Local HPROM ( $N_c = 8$ )     | 16–26              | –         | 1 781  | 0.787                       | 0.900                       | 0.070                 | 179.7   |
| Local HQPPROM ( $N_c = 8$ )   | 10–15              | –         | 1 075  | 0.787                       | 0.890                       | 0.082                 | 151.9   |
| Local HPROM-GPR ( $N_c = 8$ ) | 6                  | 28–54     | 541    | 0.801                       | 1.156                       | 0.085                 | 147.9   |

Speedup is computed with respect to the HDM runtime,  $t_{\text{HDM}} = 751.043$  s, averaged over the three test points.

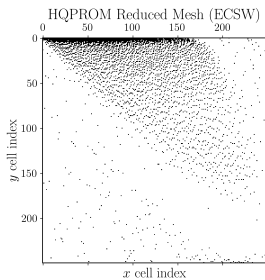
$N_e$  denotes the number of sampled ECSW elements ( $N_e = 62,500$  for HDM/full mesh).

All models are reported at a matched accuracy target of  $1\% \pm 0.2\%$  in  $\max \mathbb{RE}_{2,u}$ : for each  $N_c$  the accuracy knob is tuned to that window (pod\_tol for Local HPROM,  $\zeta_{\text{qua}}$  for Local HQPPROM,  $n$  for Local HPROM-GPR), so differences in  $n$ ,  $N_e$  and speedup are read at constant accuracy.

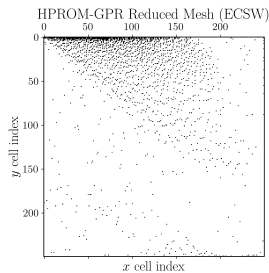
# Global Reduced Meshes (ECSW)



HPROM ( $N_e = 4,241$ )

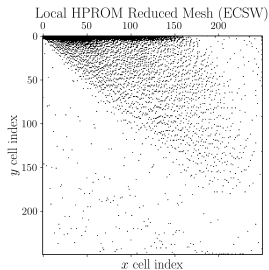


HQPROM ( $N_e = 3,460$ )

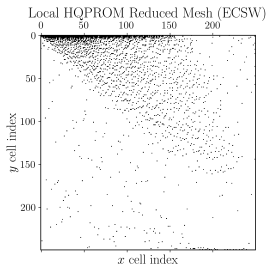


HPROM-GPR ( $N_e = 1,799$ )

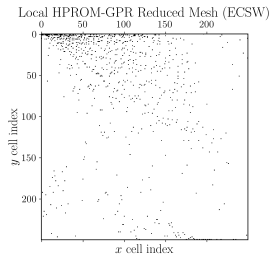
## Local Reduced Meshes (ECSW), $N_c = 3$



Local HPROM ( $N_e = 3,405$ )

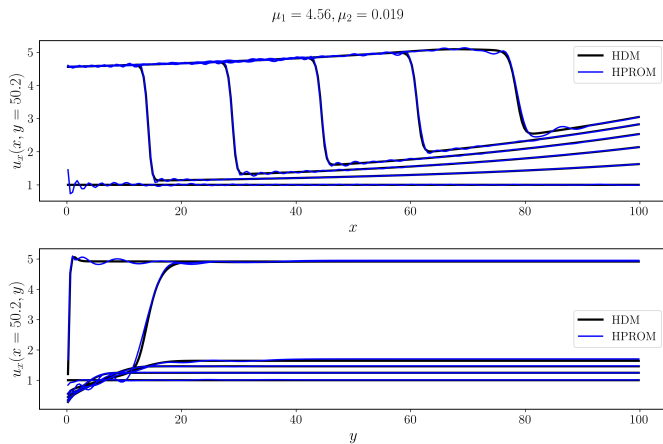


Local HQPROM ( $N_e = 2,152$ )

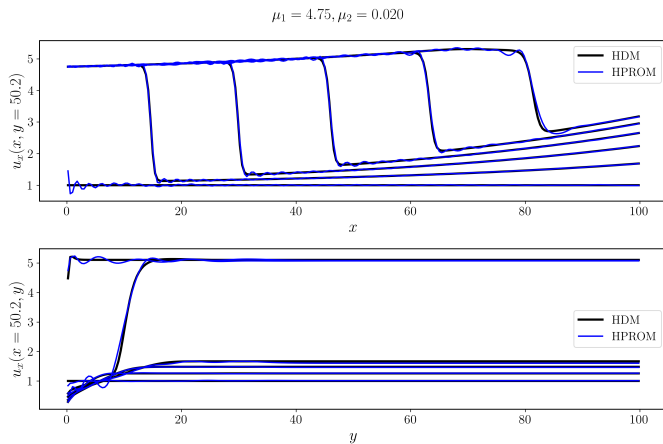


Local HPROM-GPR ( $N_e = 811$ )

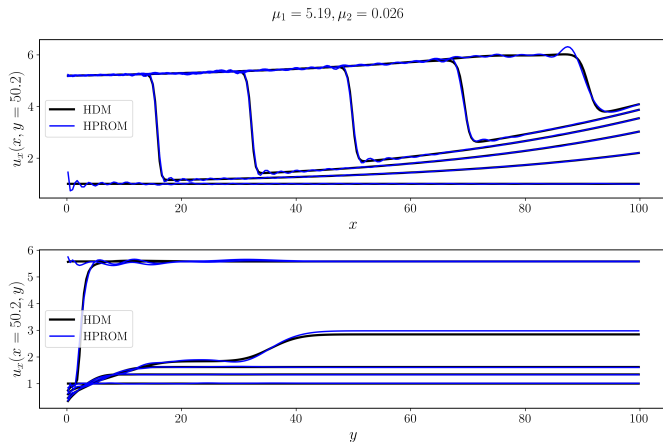
## HPROM vs HDM: $\mu = (4.56, 0.019)$



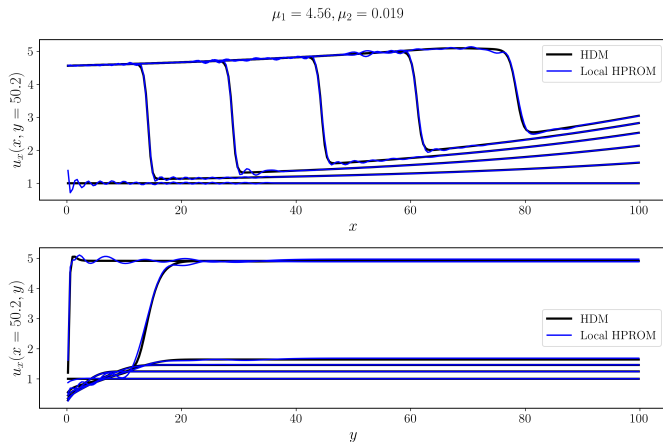
# HPROM vs HDM: $\mu = (4.75, 0.020)$



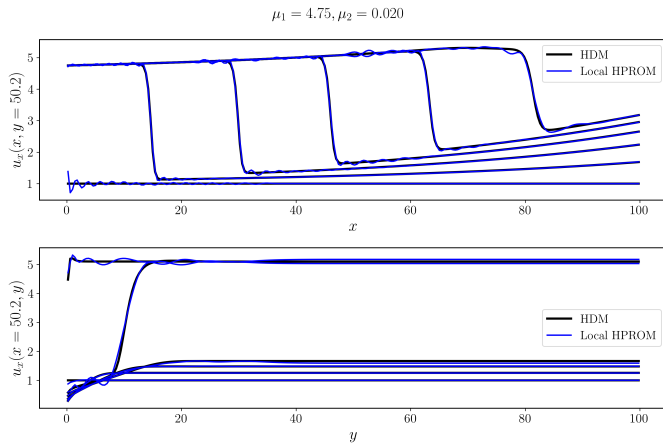
## HPROM vs HDM: $\mu = (5.19, 0.026)$



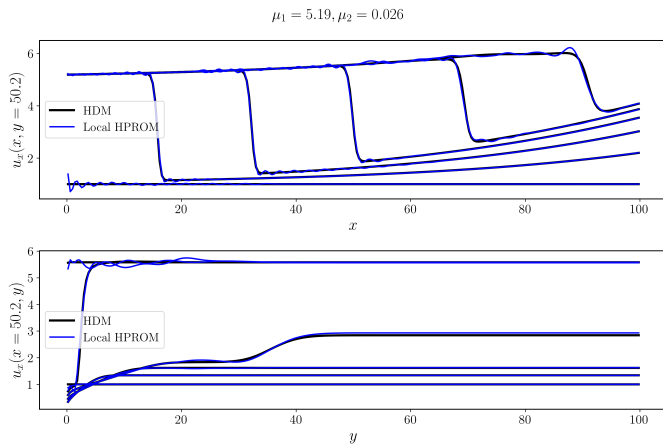
# Local HPROM ( $N_c = 3$ ) vs HDM: $\mu = (4.56, 0.019)$



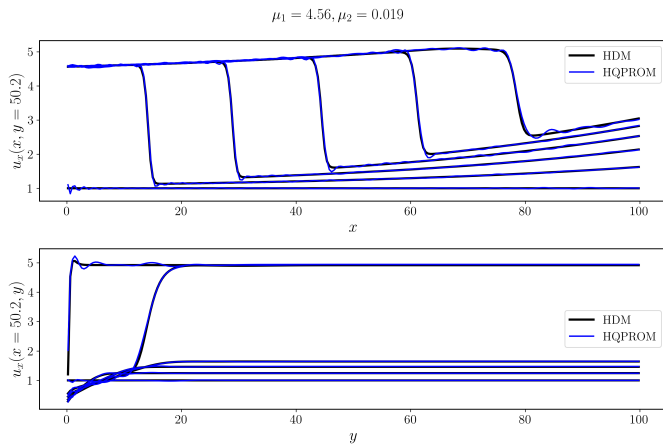
## Local HPROM ( $N_c = 3$ ) vs HDM: $\mu = (4.75, 0.020)$



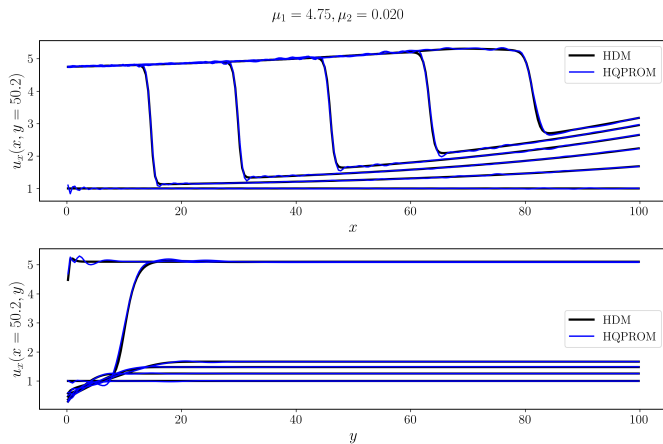
## Local HPROM ( $N_c = 3$ ) vs HDM: $\mu = (5.19, 0.026)$



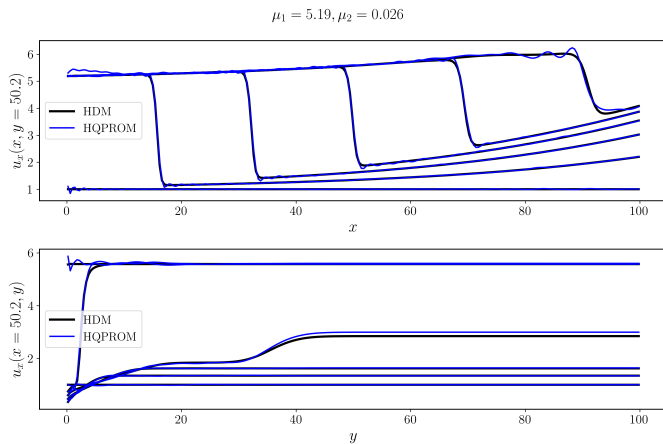
## HQPROM vs HDM: $\mu = (4.56, 0.019)$



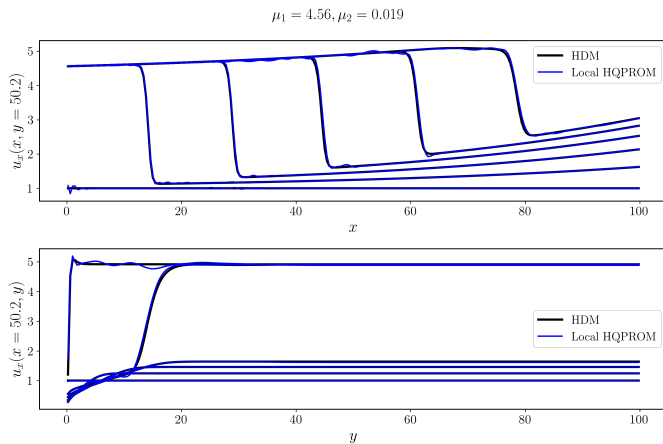
## HQPROM vs HDM: $\mu = (4.75, 0.020)$



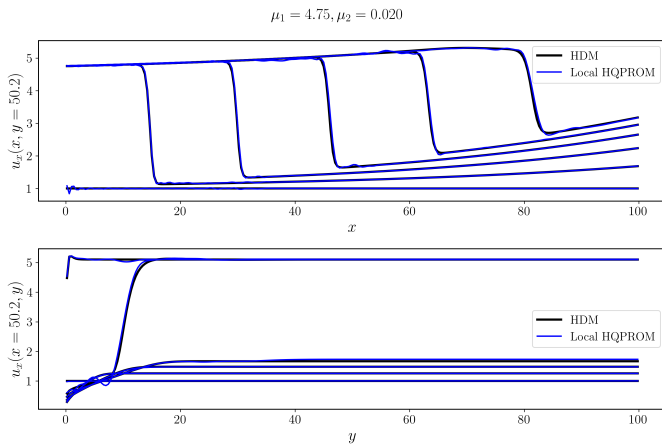
## HQPROM vs HDM: $\mu = (5.19, 0.026)$



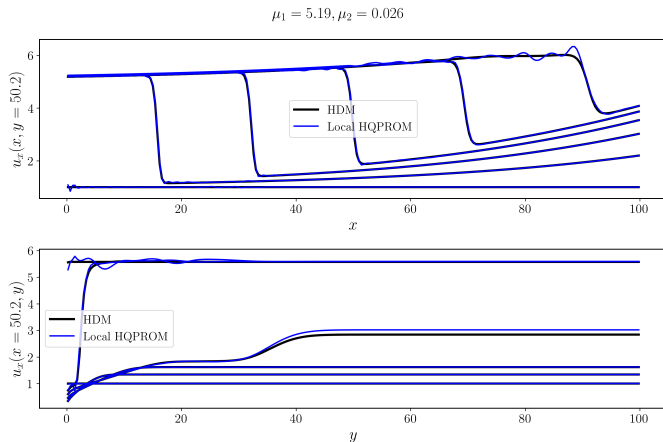
## Local HQPROM ( $N_c = 3$ ) vs HDM: $\mu = (4.56, 0.019)$



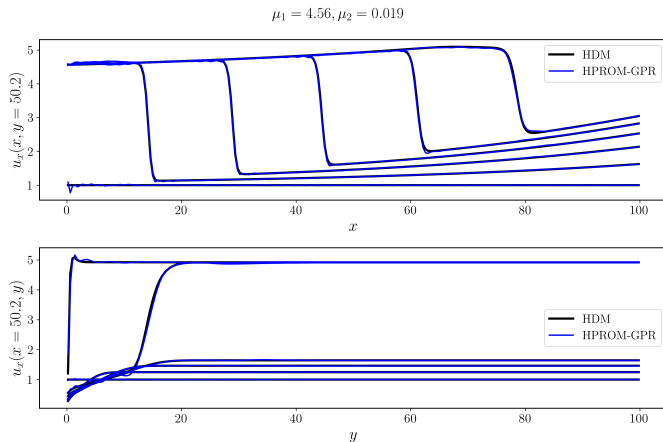
## Local HQPROM ( $N_c = 3$ ) vs HDM: $\mu = (4.75, 0.020)$



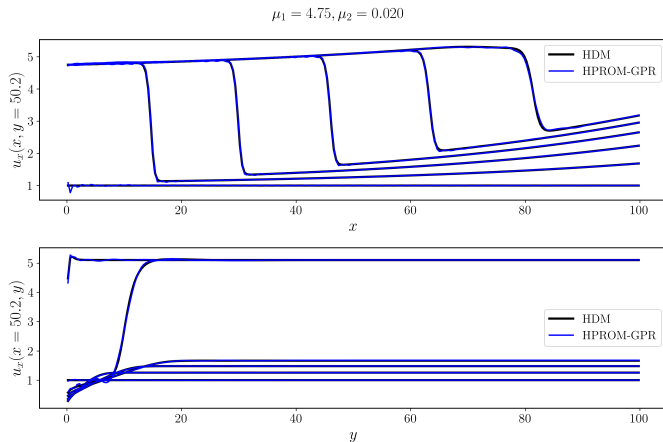
## Local HQPROM ( $N_c = 3$ ) vs HDM: $\mu = (5.19, 0.026)$



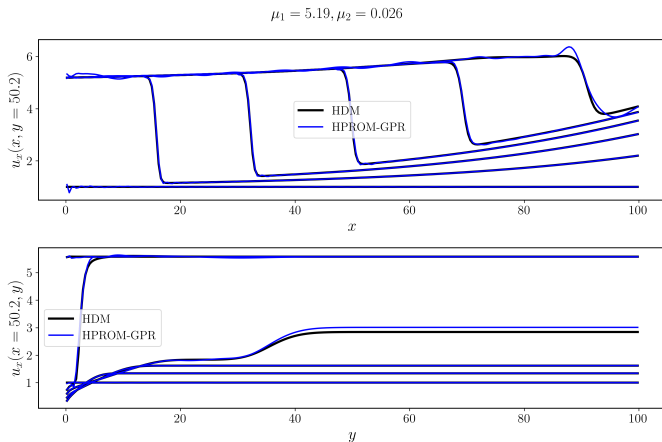
# HPROM-GPR vs HDM: $\mu = (4.56, 0.019)$



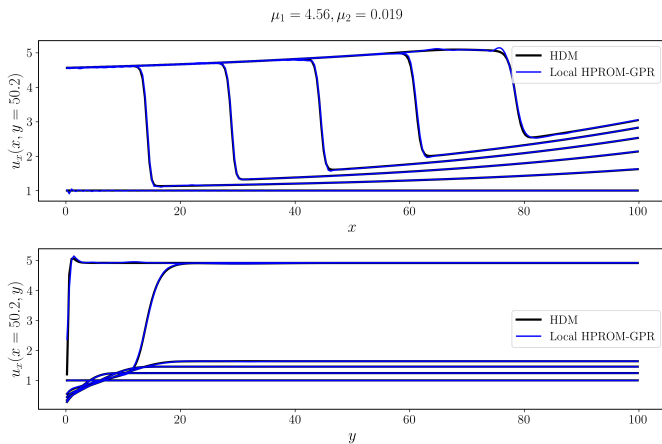
# HPROM-GPR vs HDM: $\mu = (4.75, 0.020)$



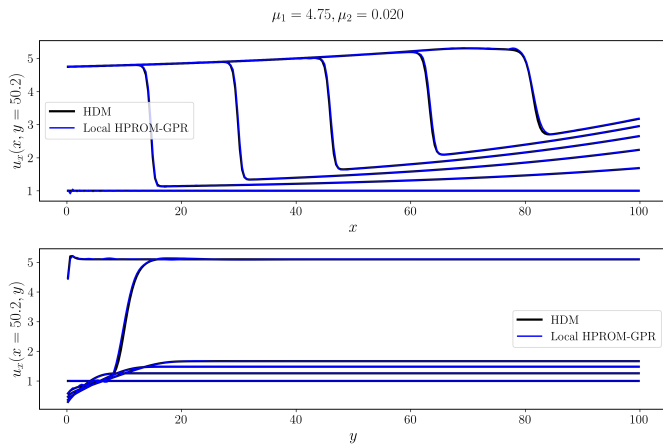
## HPROM-GPR vs HDM: $\mu = (5.19, 0.026)$



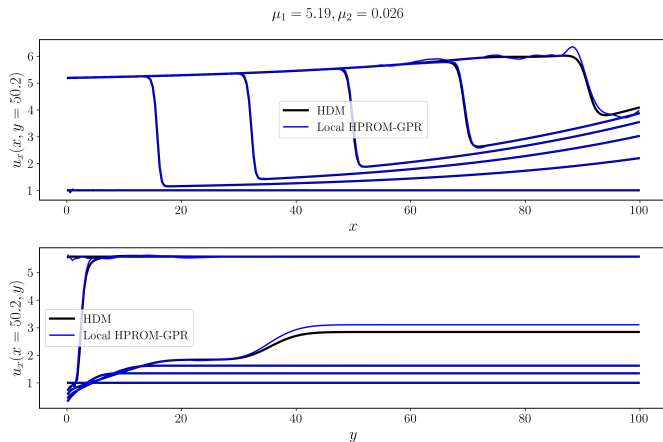
## Local HPROM-GPR ( $N_c = 3$ ) vs HDM: $\mu = (4.56, 0.019)$



## Local HPROM-GPR ( $N_c = 3$ ) vs HDM: $\mu = (4.75, 0.020)$



## Local HPROM-GPR ( $N_c = 3$ ) vs HDM: $\mu = (5.19, 0.026)$



## Appendix: Linear Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k^-} + \mathbf{V}_{k^-} \mathbf{q}_{k^-}, \quad k^+ = \arg \min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2.$$

$$\text{Let } \mathbf{a}_{\ell} := \mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k^-} \mathbf{q}_{k^-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_{\ell} + \mathbf{b}\|_2^2 = \|\mathbf{a}_{\ell}\|_2^2 + \|\mathbf{b}\|_2^2 + 2\mathbf{a}_{\ell}^T \mathbf{b}.$$

$$\|\mathbf{b}\|_2^2 = \|\mathbf{V}_{k^-} \mathbf{q}_{k^-}\|_2^2 \text{ is independent of } \ell.$$

$$\Rightarrow k^+ = \arg \min_{\ell} \left[ \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k^-} \mathbf{q}_{k^-} \right].$$

Precompute

$$\mathbf{g}_{k^-, \ell} := \mathbf{V}_{k^-}^T (\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}), \quad d_{k^-, \ell} := \|\mathbf{u}_{0,k^-} - \mathbf{u}_{c,\ell}\|_2^2,$$

which yields

$$k^+ = \arg \min_{\ell} \left( 2\mathbf{g}_{k^-, \ell}^T \mathbf{q}_{k^-} + d_{k^-, \ell} \right).$$

## Appendix: HQPROM Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-} \mathbf{q}_{k-} + \mathbf{H}_{k-} (\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k-} \mathbf{q}_{k-}, \quad \mathbf{c} := \mathbf{H}_{k-} (\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}).$$

$$k^+ = \arg \min_{\ell} \|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \arg \min_{\ell} \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2.$$

$$\|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2 = \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\mathbf{a}_\ell^T \mathbf{b} + 2\mathbf{a}_\ell^T \mathbf{c} + 2\mathbf{b}^T \mathbf{c}.$$

$$\|\mathbf{b}\|_2^2, \|\mathbf{c}\|_2^2, 2\mathbf{b}^T \mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow k^+ = \arg \min_{\ell} \left[ \|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k-} \mathbf{q}_{k-} + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \mathbf{H}_{k-} (\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}) \right].$$

Define

$$\mathbf{m}_{k-, \ell}^T (\mathbf{q} \otimes \mathbf{q}) := (\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \mathbf{H}_{k-} (\mathbf{q} \otimes \mathbf{q}),$$

then

$$k^+ = \arg \min_{\ell} \left( 2\mathbf{g}_{k-, \ell}^T \mathbf{q}_{k-} + 2\mathbf{m}_{k-, \ell}^T (\mathbf{q}_{k-} \otimes \mathbf{q}_{k-}) + d_{k-, \ell} \right).$$

## Appendix: HPRM-GPR Selector Expansion

$$\mathbf{u}^s = \mathbf{u}_{0,k-} + \mathbf{V}_{k-} \mathbf{q}_{k-} + \bar{\mathbf{V}}_{k-} \bar{\mathbf{q}}_{k-}, \quad \bar{\mathbf{q}}_{k-} = \mathcal{N}_{\text{GPR},k-}(\mathbf{q}_{k-}).$$

$$\text{Let } \mathbf{a}_\ell := \mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}, \quad \mathbf{b} := \mathbf{V}_{k-} \mathbf{q}_{k-}, \quad \mathbf{c} := \bar{\mathbf{V}}_{k-} \bar{\mathbf{q}}_{k-}.$$

$$\|\mathbf{u}^s - \mathbf{u}_{c,\ell}\|_2^2 = \|\mathbf{a}_\ell + \mathbf{b} + \mathbf{c}\|_2^2$$

$$= \|\mathbf{a}_\ell\|_2^2 + \|\mathbf{b}\|_2^2 + \|\mathbf{c}\|_2^2 + 2\mathbf{a}_\ell^T \mathbf{b} + 2\mathbf{a}_\ell^T \mathbf{c} + 2\mathbf{b}^T \mathbf{c}.$$

$$\|\mathbf{b}\|_2^2, \|\mathbf{c}\|_2^2, 2\mathbf{b}^T \mathbf{c} \text{ are independent of } \ell.$$

$$\Rightarrow k^+ = \arg \min_{\ell} \left[ \|\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}\|_2^2 + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \mathbf{V}_{k-} \mathbf{q}_{k-} + 2(\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell})^T \bar{\mathbf{V}}_{k-} \bar{\mathbf{q}}_{k-} \right].$$

Hence, with

$$\bar{\mathbf{g}}_{k-, \ell} := \bar{\mathbf{V}}_{k-}^T (\mathbf{u}_{0,k-} - \mathbf{u}_{c,\ell}),$$

$$k^+ = \arg \min_{\ell} \left( 2\mathbf{g}_{k-, \ell}^T \mathbf{q}_{k-} + 2\bar{\mathbf{g}}_{k-, \ell}^T \bar{\mathbf{q}}_{k-} + d_{k-, \ell} \right).$$